

The Choi-Jamiołkowski Isomorphism: You’re Doing It Wrong!

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As the dear departed [Quantum Pontiff](#) used to say: [New Paper Dance!](#) I am pretty happy that this one has finally been posted because it is my first arXiv paper since I returned to work, and also because it has gone through more rewrites than Spiderman: The Musical.

What is the paper about, I hear you ask? Well, mathematically, it is about an extremely simple linear algebra trick called the Choi-Jamiołkowski isomorphism. This is actually two different results: the Choi isomorphism and the Jamiołkowski isomorphism, but people have a habit of lumping them together. This trick is so extremely well-known to quantum information theorists that it is not even funny. One of the main points of the paper is that you should think about what the isomorphism means physically in a new way. Hence the “you’re doing it wrong” in the post title.

FIRST LEVEL ISOMORPHISMS

For the uninitiated, here is the simplest way of describing the Choi isomorphism in a single equation:

$$|j\rangle\langle k| \equiv |j\rangle \otimes |k\rangle,$$

i.e. the isomorphism works by turning a bra into a ket. The thing on the left is an operator on a Hilbert space $\mathcal{H}$ and the thing on the right is a vector in $\mathcal{H} \otimes \mathcal{H}$, so the isomorphism says that $\mathcal{L}(\mathcal{H}) \equiv \mathcal{H} \otimes \mathcal{H}$, where $\mathcal{L}(\mathcal{H})$ is the space of linear operators on $\mathcal{H}$.

Here is how it works in general. If you have an operator U then you can pick a basis for $\mathcal{H}$ and write U in this basis as

$$U = \sum_{j,k} U_{j,k} |j\rangle\langle k|,$$

where $U_{j,k} = \langle j|U|k\rangle$. Then you just extend the above construction by linearity and write down a vector

$$|\Phi_U\rangle = \sum_{j,k} U_{j,k} |j\rangle \otimes |k\rangle.$$

It is pretty obvious that we can go in the other direction as well, starting with a vector on $\mathcal{H} \otimes \mathcal{H}$, we can write it out in a product basis, turn the second ket into a bra, and then we have an operator.

So far, this is all pretty trivial linear algebra, but when we think about what this means physically it is pretty weird. One of the things that is represented by an operator in quantum theory is dynamics, in particular a unitary operator represents the dynamics of a closed system for a discrete time-step. One of the things that is represented by a vector on a tensor product Hilbert space is a pure state of a bipartite system. It is fairly easy to see that (up to normalization) unitary operators get mapped to maximally entangled states under the isomorphism, so, in some sense, a maximally entangled state is “the same thing” as a unitary operator. This is weird because there are some things that make sense for dynamical operators that don’t seem to make sense for states and vice-versa. For example, dynamics can be composed. If U represents the dynamics from t_0 to t_1 and V represents the dynamics from t_1 to t_2 , then the dynamics from t_0 to t_2 is represented by the product VU . Using the isomorphism, we can define a composition for states, but what on earth does this mean?

Before getting on to that, let us briefly pause to consider the Jamiolkowski version of the isomorphism. The Choi isomorphism is basis dependent. You get a slightly different state if you write down the operator in a different basis. To make things basis independent, we replace $\mathcal{H} \otimes \mathcal{H}$ by $\mathcal{H} \otimes \mathcal{H}^*$. $\mathcal{H}^*$ denotes the dual space to $\mathcal{H}$, i.e. it is the space of bras instead of the space of kets. In Dirac notation, the Jamiolkowski isomorphism looks pretty trivial. It says

$$|j\rangle \langle k| \equiv |j\rangle \otimes \langle k|.$$

This is axiomatic in Dirac notation, because we always assume that tensor product symbols can be omitted without changing anything. However, this version of the isomorphism is going to become important later.

CONVENTIONAL INTERPRETATION: GATE TELEPORTATION

In quantum information, the Choi isomorphism is usually interpreted in terms of “gate teleportation”. To understand this, we first reformulate the isomorphism slightly. Let $|\Phi^+\rangle_{AA'} = \sum_j |jj\rangle_{AA'}$, where A and A' are quantum systems with Hilbert spaces of the same dimension. The vectors $|j\rangle$ form a preferred basis, and this is the basis in which the Choi isomorphism is going to be defined. Note that $|\Phi^+\rangle_{AA'}$ is an (unnormalized) maximally entangled state. It is easy to check that the isomorphism can now be reformulated as

$$|\Phi_U\rangle_{AA'} = I_A \otimes U'_A |\Phi^+\rangle_{AA'},$$

where I_A is the identity operator on system A . The reverse direction of the isomorphism is given by

$$U_A |\psi\rangle \langle \psi|_A U_A^\dagger = \langle \Phi^+|_{A'A''} (|\psi\rangle \langle \psi|_{A''} \otimes |\Phi_U\rangle \langle \Phi_U|_{A'A}) |\Phi^+\rangle_{A'A''},$$

where A'' is yet another quantum system with the same Hilbert space as A .

Now let’s think about the physical interpretation of the reverse direction of the isomorphism. Suppose that U is the identity. In that case, $|\Phi_U\rangle = |\Phi^+\rangle$ and the reverse direction of the isomorphism is easily recognized as the expression for the output of the teleportation protocol when the $|\Phi^+\rangle$ outcome is obtained in the Bell measurement. It says that $|\psi\rangle$ gets teleported from A'' to A . Of course, this outcome only occurs some of the time, with probability $1/d$, where d is the dimension of the Hilbert space of A , a fact that is obscured by our decision to use an unnormalized version of $|\Phi^+\rangle$.

Now, if we let U be a nontrivial unitary operator then the reverse direction of the isomorphism says something more interesting. If we use the state $|\Phi_U\rangle$ rather than $|\Phi^+\rangle$ as our resource state in the teleportation protocol, then, upon

obtaining the $|\Phi^+\rangle$ outcome in the Bell measurement, the output of the protocol will not simply be the input state $|\psi\rangle$, but it will be that state with the unitary U applied to it. This is called “gate teleportation”. It has many uses in quantum computing. For example, in linear optics implementations, it is impossible to perform every gate in a universal set with 100% probability. To avoid damaging your precious computational state, you can apply the indeterministic gates to half of a maximally entangled state and keep doing so until you get one that succeeds. Then you can teleport your computational state using the resulting state as a resource and end up applying the gate that you wanted. This allows you to use indeterministic gates without having to restart the computation from the beginning every time one of these gates fails.

Using this interpretation of the isomorphism, we can also come up with a physical interpretation of the composition of two states. It is basically a generalization of entanglement swapping. If you take $|\Phi_U\rangle$ and $|\Phi_V\rangle$ and perform a Bell measurement across the output system of the first and the input system of the second then, upon obtaining the $|\Phi^+\rangle$ outcome, you will have the state $|\Phi_{UV}\rangle$. In this way, you can perform your entire computational circuit in advance, before you have access to the input state, and then just teleport your input state into the output register as the final step.

In this way, the Choi isomorphism leads to a correspondence between a whole host of protocols involving gates and protocols involving entangled states. We can also define interesting properties of operations, such as the entanglement of an operation, in terms of the states that they correspond to. We then use the isomorphism to give a physical meaning to these properties in terms of gate teleportation. However, one weak point of the correspondence is that it transforms something deterministic; the application of a unitary operation; into something indeterministic; getting the $|\Phi^+\rangle$ outcome in a Bell measurement. Unlike the teleportation protocol, gate teleportation cannot be made deterministic by applying correction operations for the other outcomes, at least not if we want these corrections to be independent of U . The states you get for the other outcomes involve nasty things like U^* , U^T , $U^\dagger$ applied to $|\psi\rangle$, depending on exactly how you construct the Bell basis, e.g. choice of phases. These can typically not be corrected without applying U . In particular, that would screw things up in the linear optics application wherein U can only be implemented non-deterministically.

Before turning to our alternative interpretation of Choi-Jamiołkowski, let's generalize things a bit.

SECOND LEVEL ISOMORPHISMS

In quantum theory we don't just have pure states, but also mixed states that arise if you have uncertainty about which state was prepared, or if you ignore a subsystem of a larger system that is in a pure state. These are described by positive, trace-one, operators, denoted ρ , called density operators. Similarly, dynamics does not have to be unitary. For example, we might bring in an extra system, interact them unitarily, and then trace out the extra system. These are described by Completely-Positive, Trace-Preserving (CPT) maps, denoted $\mathcal{E}$. These are linear maps that act on the space of operators, i.e. they are operators on the space of operators, and are often called superoperators.

Now, the set of operators on a Hilbert space is itself a Hilbert space with inner product $\langle N, M \rangle = \text{Tr}(N^\dagger M)$. Thus, we can apply Choi-Jamiołkowski on this space to define a correspondence between superoperators and operators on the tensor product. We can do this in terms of an orthonormal operator basis with respect to the trace inner product, but it is easier to just give the teleportation version of the isomorphism. We will also generalize slightly to allow for the possibility that the input and output spaces of our CPT map may be different, i.e. it may involve discarding a subsystem of the system we started with, or bringing in extra ancillary systems.

Starting with a CPT map $\mathcal{E}_{B|A} : \mathcal{L}(\mathcal{H}_A) \rightarrow \mathcal{L}(\mathcal{H}_B)$ from system A to system B , we can define an operator on $\mathcal{H}_A \otimes \mathcal{H}_B$ via

$$\rho_{AB} = \mathcal{E}_{B|A} \otimes \mathcal{I}_A \left(|\Phi^+\rangle \langle \Phi^+|_{AA'} \right),$$

where $\mathcal{I}_A$ is the identity superoperator. This is a positive operator, but it is not quite a density operator as it satisfies $\text{Tr}_B(\rho_{AB}) = I_A$, which implies that $\text{Tr}_{AB}(\rho_{AB}) = d$ rather than $\text{Tr}_{AB}(\rho_{AB}) = 1$. This is analogous to using unnormalized states in the pure-state case. The reverse direction of the isomorphism is then given by

$$\mathcal{E}_{B|A}(\sigma_A) = \langle \Phi^+|_{A'A} \sigma_{A'} \otimes \rho_{AB} |\Phi^+\rangle_{A'A}.$$

This has the same interpretation in terms of gate teleportation (or rather CPT-map teleportation) as before.

The Jamiolkowski version of this isomorphism is given by

$$\varrho_{AB} = \mathcal{E}_{B|A} \otimes \mathcal{I}_A \left(|\Phi^+\rangle \langle \Phi^+|_{AA'}^{T_A} \right),$$

where T_A denotes the partial transpose in the basis used to define $|\Phi^+\rangle$. Although it is not obvious from this formula, this operator is independent of the choice of basis, as $|\Phi^+\rangle \langle \Phi^+|_{AA'}^{T_A}$ is actually the same operator for any choice of basis. I'll keep the reverse direction of the isomorphism a secret for now, as it would give a strong hint towards the punchline of this blog post.

PROBABILITY THEORY

I now want to give an alternative way of thinking about the isomorphism, in particular the Jamiolkowski version, that is in many ways conceptually clearer than the gate teleportation interpretation. The starting point is the idea that quantum theory can be viewed as a noncommutative generalization of classical probability theory. This idea goes back at least to von Neumann, and is at the root of our thinking in quantum information theory, particularly in quantum Shannon theory. The basic idea of the generalization is that that probability distributions $P(X)$ get mapped to density operators ρ_A and sums over variables become partial traces. Therefore, let's start by thinking about whether there is a classical analog of the isomorphism, and, if so, what its interpretation is.

Suppose we have two random variables, X and Y . We can define a conditional probability distribution of Y given X , $P(Y|X)$, as a positive function of the two variables that satisfies $\sum_Y P(Y|X) = 1$ independently of the value of X . Given a conditional probability distribution and a marginal distribution, $P(X)$, for X , we can define a joint distribution via

$$P(X, Y) = P(Y|X)P(X).$$

Conversely, given a joint distribution $P(X, Y)$, we can find the marginal $P(X) = \sum_Y P(X, Y)$ and then define a conditional distribution

$$P(Y|X) = \frac{P(X, Y)}{P(X)}.$$

Note, I'm going to ignore the ambiguities in this formula that occur when $P(X)$ is zero for some values of X .

Now, suppose that X and Y are the input and output of a classical channel. I now want to think of the probability distribution of Y as being determined by a stochastic map $\Gamma_{Y|X}$ from the space of probability distributions over X to the space of probability distributions over Y . Since $P(Y) = \sum_X P(X, Y)$, this has to be given by

$$P(Y) = \Gamma_{Y|X}(P(X)) = \sum_X P(Y|X)P(X),$$

or

$$\Gamma_{Y|X}(\cdot) = \sum_X P(Y|X)(\cdot)$$

What we have here is a correspondence between a positive function of two variables — the conditional probability distribution — and a linear map that acts on the space of probability distributions — the stochastic map. This looks analogous to the Choi-Jamiolkowski isomorphism, except that, instead of a joint probability distribution, which would be analogous to a quantum state, we have a conditional probability distribution. This suggests that we made a mistake in thinking of the operator in the Choi isomorphism as a state. Maybe it is something more like a conditional state.

CONDITIONAL STATES

Let's just plunge in and make a definition of a conditional state, and then see how it makes sense of the Jamiolkowski isomorphism. For two quantum systems, A and B , a conditional state of B given A is defined to be a positive operator $\rho_{B|A}$ on $\mathcal{H}_A \otimes \mathcal{H}_B$ that satisfies

$$\text{Tr}_B(\rho_{B|A}) = I_A.$$

This is supposed to be analogous to the condition $\sum_Y P(Y|X) = 1$. Notice that this is exactly how the operators that are Choi-isomorphic to CPT maps are normalized.

Given a conditional state, $\rho_{B|A}$, and a reduced state ρ_A , I can define a joint state via

$$\rho_{AB} = \sqrt{\rho_A} \rho_{B|A} \sqrt{\rho_A},$$

where I have suppressed the implicit $\otimes I_B$ required to make the products well defined. The conjugation by the square root ensures that ρ_{AB} is positive, and it is easy to check that $\text{Tr}_{AB}(\rho_{AB}) = 1$.

Conversely, given a joint state, I can find its reduced state $\rho_A = \text{Tr}_B(\rho_{AB})$ and then define the conditional state

$$\rho_{B|A} = \sqrt{\rho_A^{-1}} \rho_{AB} \sqrt{\rho_A^{-1}},$$

where I am going to ignore cases in which ρ_A has any zero eigenvalues so that the inverse is well-defined (this is

no different from ignoring the division by zero in the classical case).

Now, suppose you are given ρ_A and you want to know what ρ_B should be. Is there a linear map that tells you how to do this, analogous to the stochastic map $\Gamma_{Y|X}$ in the classical case? The answer is obviously yes. We can define a map $\mathfrak{E}_{B|A} : \mathcal{L}(\mathcal{H}_A) \rightarrow \mathcal{L}(\mathcal{H}_B)$ via

$$\mathfrak{E}_{B|A}(\rho_A) = \text{Tr}_A(\rho_{B|A}\rho_A),$$

where we have used the cyclic property of the trace to combine the $\sqrt{\rho_A}$ terms, or

$$\mathfrak{E}_{B|A}(\cdot) = \text{Tr}_A(\rho_{B|A}(\cdot)).$$

The map $\mathfrak{E}_{B|A}$ so defined is just the Jamiolkowski isomorphic map to $\rho_{B|A}$ and the above equation gives the reverse direction of the Jamiolkowski isomorphism that I was being secretive about earlier.

The punchline is that the Choi-Jamiolkowski isomorphism should not be thought of as a mapping between quantum states and quantum operations, but rather as a mapping between conditional quantum states and quantum operations. It is no more surprising than the fact that classical stochastic maps are determined by conditional probability distributions. If you think of it in this way, then your approach to quantum information will become conceptually simpler a lot of ways. These ways are discussed in detail in the paper.

CAUSAL CONDITIONAL STATES

There is a subtlety that I have glossed over so far that I'd like to end with. The map $\mathfrak{E}_{B|A}$ is not actually completely positive, which is why I did not denote it $\mathcal{E}_{B|A}$, but when preceded by a transpose on A it defines a completely positive map. This is because the Jamiolkowski isomorphism is defined in terms of the partial transpose of the maximally entangled state. Also, so far I have been talking about two distinct quantum systems that exist at the same time, whereas in the classical case, I talked about the input and output of a classical channel. A quantum channel is given by a CPT map $\mathcal{E}_{B|A}$ and its Jamiolkowski representation would be

$$\mathcal{E}_{B|A}(\rho_A) = \text{Tr}_A(\varrho_{B|A}\rho_A),$$

where $\varrho_{B|A}$ is the partial transpose over A of a positive operator and it satisfies $\text{Tr}_B(\varrho_{B|A}) = I_A$. This is the appropriate notion of a conditional state in the causal scenario, where you are talking about the input and output of a quantum channel rather than two systems at the same time. The two types of conditional state are related by a partial transpose.

Despite this difference, a good deal of unification is achieved between the way in which acausally related (two subsystems) and causally related (input and output of channels) degrees of freedom are described in this framework. For example, we can define a "causal joint state" as

$$\varrho_{AB} = \sqrt{\rho_A}\varrho_{B|A}\sqrt{\rho_A},$$

where ρ_A is the input state to the channel and $\varrho_{B|A}$ is the Jamiolkowski isomorphic map to the CPT map. This unification is another main theme of the paper, and allows a quantum version of Bayes' theorem to be defined that is independent of the causal scenario.

THE WONDERFUL WORLD OF CONDITIONAL STATES

To end with, here is a list of some things that become conceptually simpler in the conditional states formalism developed in the paper:

- The Born rule, ensemble averaging, and quantum dynamics are all just instances of a quantum analog of the formula $P(Y) = \sum_X P(Y|X)P(X)$.
- The Heisenberg picture is just a quantum analog of $P(Z|X) = \sum_Y P(Z|Y)P(Y|X)$.
- The relationship between prediction and retrodiction (inferences about the past) in quantum theory is given by the quantum Bayes' theorem.
- The formula for the set of states that a system can be 'steered' to by making measurements on a remote system, as in EPR-type experiments, is just an application of the quantum Bayes' theorem.

If this has whet your appetite, then this and much more can be found in [the paper](#).

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